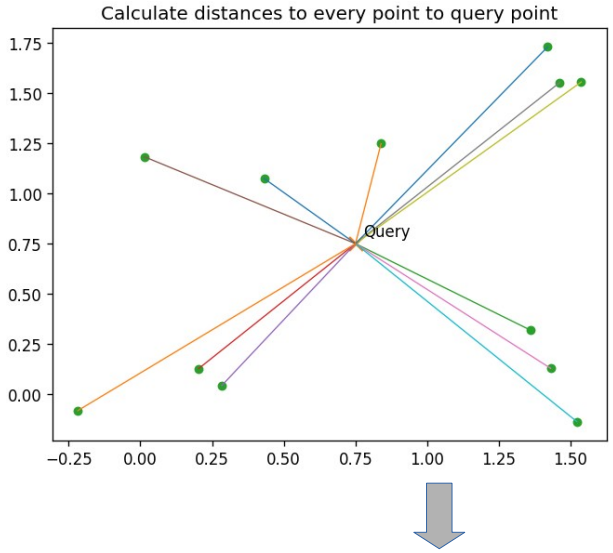
Similarity Searchin Vector Databases: kNN, IVF, HNSW

An overview

K Nearest Neighbors (kNN)

- Finds the k closest points to a query
- By computing distances to all n points
- Cost: ~O(n) per query → slow at scale

kNN- algorithm



Distance
1.275413
0.847395
0.829449
0.746751
0.921099
1.174519
1.072903
1.187165
1.125681
0.454620
0.853182
0.505817

Distances

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1.072903
1.125681
1.174519
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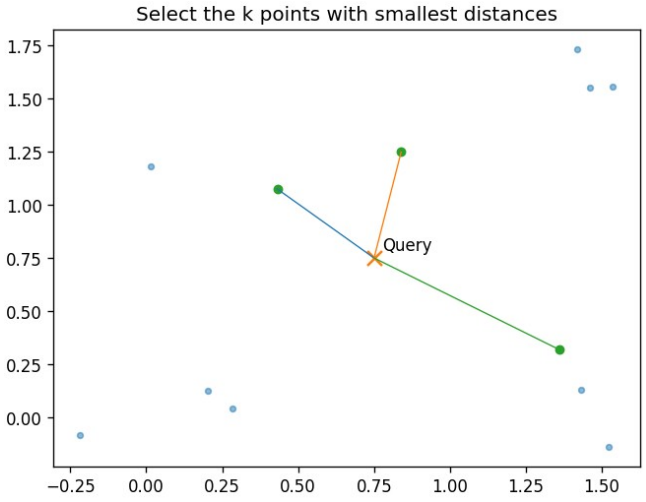
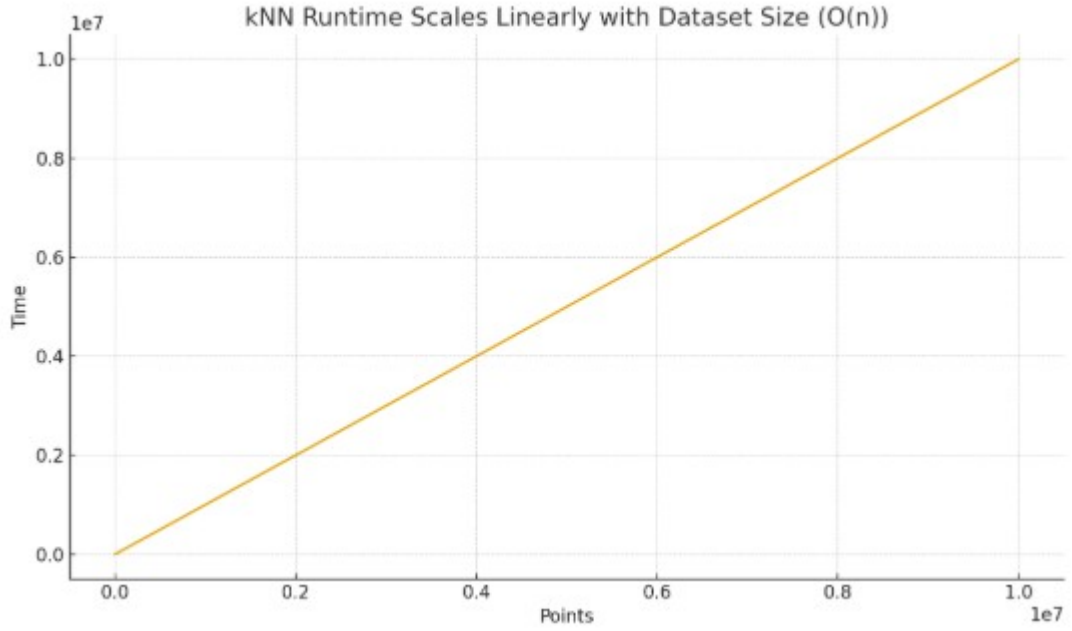
Sorted Distances

Select k closest



Why kNN Doesn’t Scale

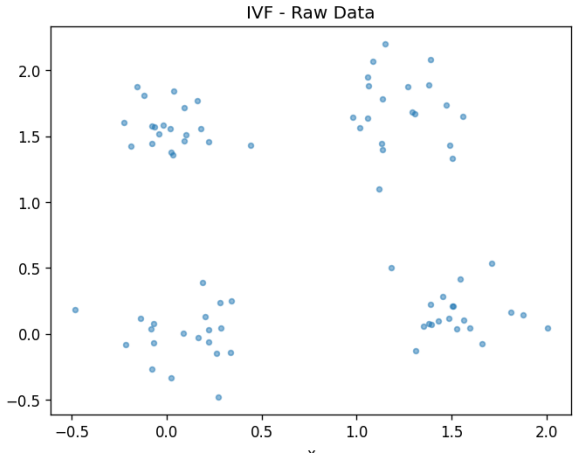
- Per-query time grows linearly with data size: exact kNN must compare a query to every point → O(n) work per query
- latency explodes as n grows.

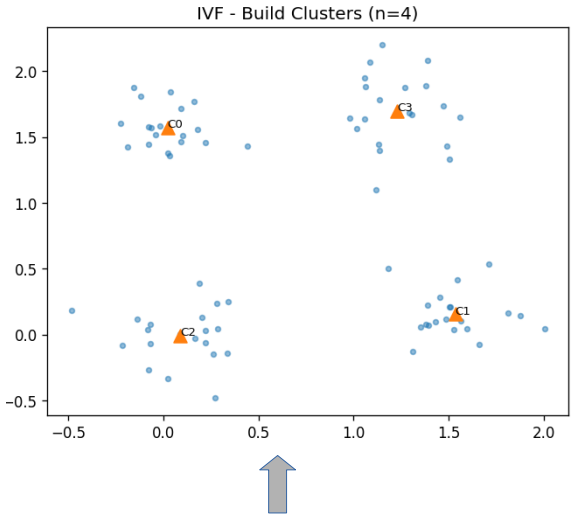


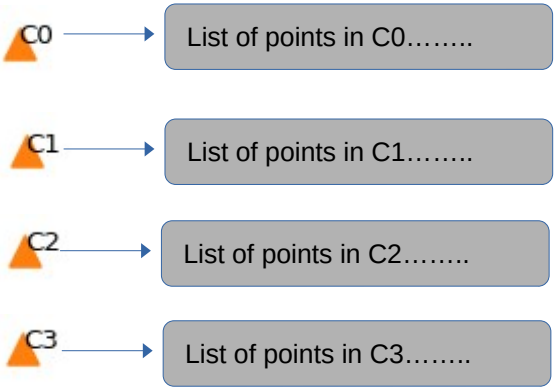
IVF: Inverted File Index — Idea

- Partition space into n clusters
  - Rule of thumb: 1000 clusters for 10000000 points -
- At query time probe only k relevant clusters

IVF: Build Clusters





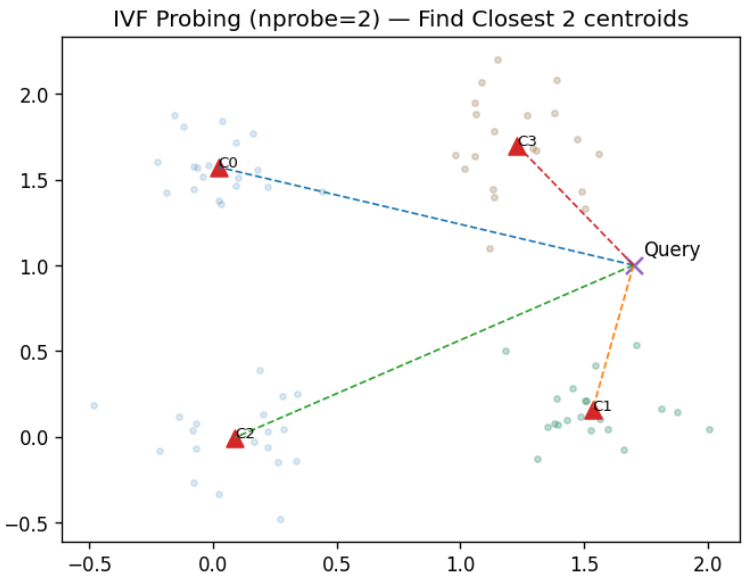


This is an iterative process (KMeans)

1) Randomly place 4 centroids
2) Assign each point to its closest cluster
3) Move centroid to center of its points
4) Go to 2 until no points change cluster

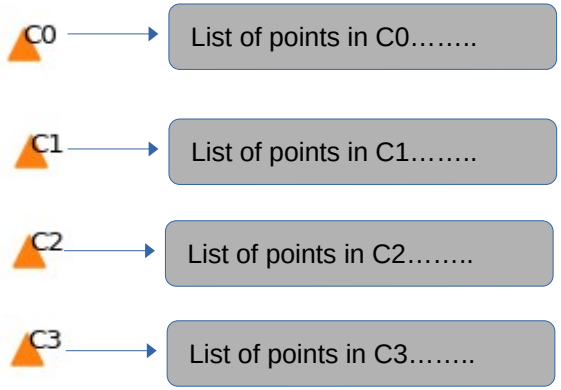
Have list of all points in each cluster

IVF: Query time

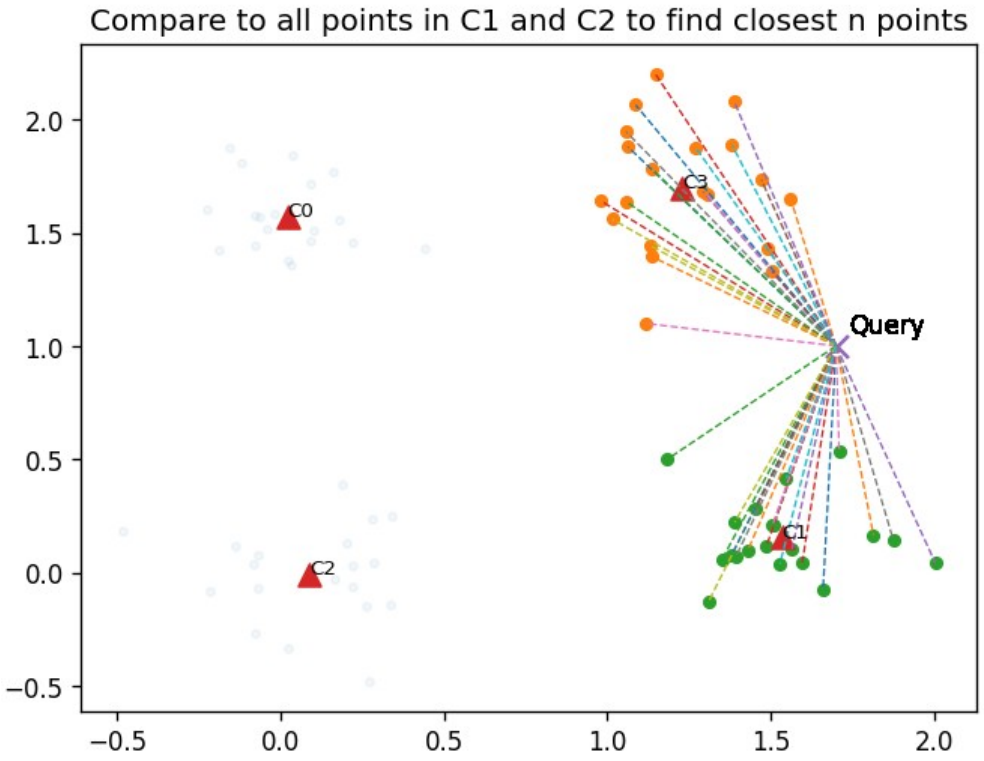


C1 and C3 are closest 2 centroids

Use the Lists for C1 and C3 to get all the points to compare to



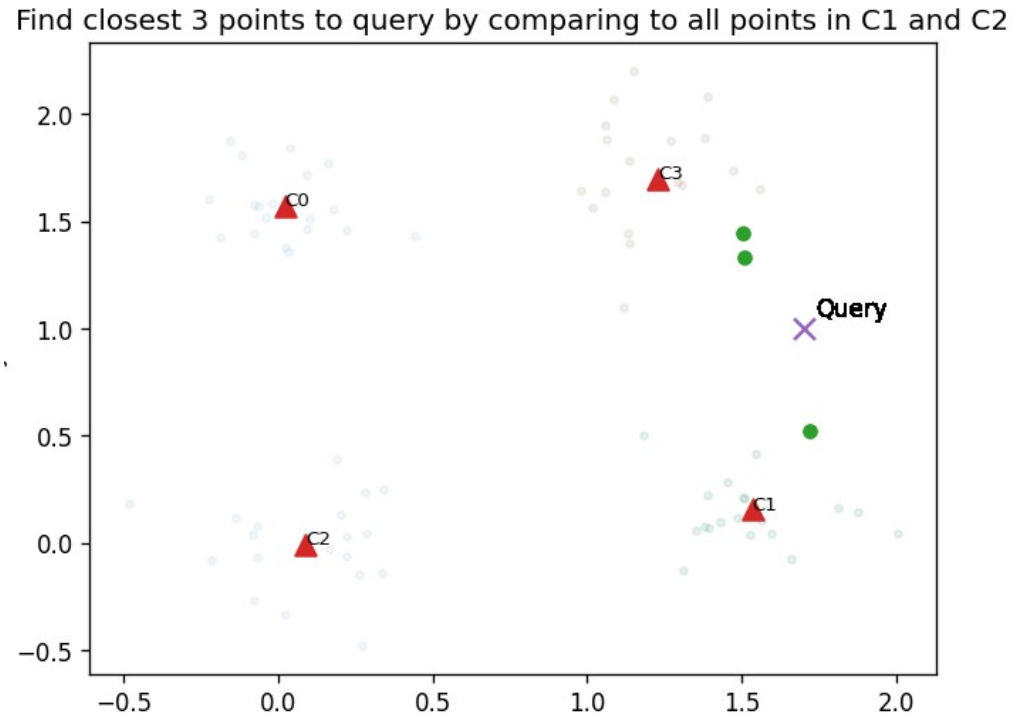
IVF: Query time



Just like kNN

- Get distances
- Sort distances

IVF: Query time

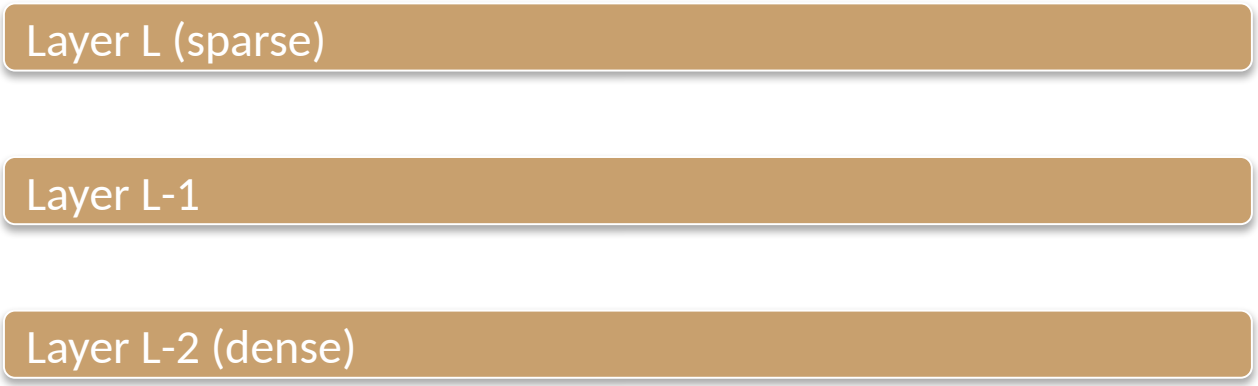


Just like kNN

- Get distances
- Sort distances
- Select closest n points

HNSW: Small-World Graph — Idea

Multi-layer graph; greedy search from top layer, descending to denser layers.



HNSW: Mechanics & Tuning

- Build: M = max neighbors; efConstruction controls build quality/time
- Query: efSearch controls recall vs latency
- Trade-offs: ↑ M, ↑ efSearch → ↑ recall & memory/latency

Quick Comparison

Aspect	kNN (Exact)	IVF	HNSW
Query time	Highest (O(n))	Low, tunable (nprobe - #clusters)	Very low, tunable (efSearch)
Build time	None	Moderate (k-means)	High (graph build)
Memory	Data only	Data + centroids	Data + graph links (higher)

- kNN will find exact match
- IVF and HNSW may not

Summary

- kNN is exact but slow for large n
- IVF: partition + probe a few lists (nlist, nprobe)
- HNSW: layered graph, fast greedy search (M, efSearch)